Deep Learning Portfolio Examination

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Part - 1: Stabilizing the Existing Codebase

Performance Report

Each of the four recovered diagnostic image profiles (cells, chest, lesions, orgs) was benchmarked against all three architectures (AlexNet, VGG16, ResNet18) using the reconstructed, config-driven pipeline. The complete per-epoch training and validation logs and the final held-out test metrics are reproduced below for every dataset-model permutation.

Dataset Used – cells

Blood cell microscopy (8 classes, RGB) | Expected minimum test accuracy: 90%

1. cells – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 3
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 8
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.9301 - Train Acc: 63.56% | Val Loss: 3.1397 - Val Acc: 36.36%
Epoch [02/15] | Train Loss: 0.4248 - Train Acc: 84.58% | Val Loss: 0.9293 - Val Acc: 68.25%
Epoch [03/15] | Train Loss: 0.3071 - Train Acc: 89.87% | Val Loss: 0.3447 - Val Acc: 88.59%
Epoch [04/15] | Train Loss: 0.2565 - Train Acc: 91.68% | Val Loss: 0.5073 - Val Acc: 79.81%
Epoch [05/15] | Train Loss: 0.2163 - Train Acc: 93.25% | Val Loss: 2.3361 - Val Acc: 57.94%
Epoch [06/15] | Train Loss: 0.1875 - Train Acc: 93.78% | Val Loss: 0.4023 - Val Acc: 86.32%
Epoch [07/15] | Train Loss: 0.1736 - Train Acc: 94.26% | Val Loss: 1.6407 - Val Acc: 64.01%
Epoch [08/15] | Train Loss: 0.1862 - Train Acc: 94.21% | Val Loss: 0.8748 - Val Acc: 69.86%
Epoch [09/15] | Train Loss: 0.1644 - Train Acc: 94.78% | Val Loss: 0.2744 - Val Acc: 89.83%
Epoch [10/15] | Train Loss: 0.1454 - Train Acc: 95.27% | Val Loss: 0.2365 - Val Acc: 92.90%
Epoch [11/15] | Train Loss: 0.1482 - Train Acc: 95.17% | Val Loss: 0.2147 - Val Acc: 93.12%
Epoch [12/15] | Train Loss: 0.1422 - Train Acc: 95.55% | Val Loss: 0.2316 - Val Acc: 92.47%
Epoch [13/15] | Train Loss: 0.1256 - Train Acc: 95.94% | Val Loss: 0.3447 - Val Acc: 91.15%
Epoch [14/15] | Train Loss: 0.1046 - Train Acc: 96.76% | Val Loss: 0.2586 - Val Acc: 92.90%
Epoch [15/15] | Train Loss: 0.0985 - Train Acc: 96.85% | Val Loss: 0.2213 - Val Acc: 93.64%
Best checkpoint saved: best_model/cells_AlexNet.pth (epoch=15, val_acc=93.64%)
Training Complete!
```

Final Test Results (AlexNet on cells): Accuracy = 95.15% | Precision (macro) = 0.9474 | Recall (macro) = 0.9443 | Macro F1 = 0.9445

2. cells – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 3
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 8
OPTIMIZER.................... Adam
```

```
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.7807 - Train Acc: 70.35% | Val Loss: 0.5020 - Val Acc: 81.57%
Epoch [02/15] | Train Loss: 0.4778 - Train Acc: 83.65% | Val Loss: 0.7367 - Val Acc: 78.05%
Epoch [03/15] | Train Loss: 0.3864 - Train Acc: 87.65% | Val Loss: 0.6123 - Val Acc: 77.76%
Epoch [04/15] | Train Loss: 0.3115 - Train Acc: 90.32% | Val Loss: 0.6647 - Val Acc: 75.20%
Epoch [05/15] | Train Loss: 0.3170 - Train Acc: 90.10% | Val Loss: 0.6265 - Val Acc: 80.03%
Epoch [06/15] | Train Loss: 0.2479 - Train Acc: 92.34% | Val Loss: 0.3569 - Val Acc: 89.39%
Epoch [07/15] | Train Loss: 0.2155 - Train Acc: 93.08% | Val Loss: 0.2055 - Val Acc: 93.93%
Epoch [08/15] | Train Loss: 0.2004 - Train Acc: 93.95% | Val Loss: 0.4448 - Val Acc: 86.83%
Epoch [09/15] | Train Loss: 0.1843 - Train Acc: 94.57% | Val Loss: 0.2220 - Val Acc: 93.05%
Epoch [10/15] | Train Loss: 0.1573 - Train Acc: 95.13% | Val Loss: 0.1483 - Val Acc: 95.68%
Epoch [11/15] | Train Loss: 0.1366 - Train Acc: 95.77% | Val Loss: 0.3248 - Val Acc: 88.66%
Epoch [12/15] | Train Loss: 0.1574 - Train Acc: 95.20% | Val Loss: 0.1593 - Val Acc: 94.51%
Epoch [13/15] | Train Loss: 0.1475 - Train Acc: 95.42% | Val Loss: 0.1970 - Val Acc: 93.34%
Epoch [14/15] | Train Loss: 0.1287 - Train Acc: 95.91% | Val Loss: 0.1637 - Val Acc: 95.25%
Epoch [15/15] | Train Loss: 0.1112 - Train Acc: 96.65% | Val Loss: 0.1890 - Val Acc: 94.15%
Best checkpoint saved: best_model/cells_VGG16.pth (epoch=10, val_acc=95.68%)
Training Complete!
```

Final Test Results (VGG16 on cells): Accuracy = 96.78% | Precision (macro) = 0.9646 | Recall (macro) = 0.9685 | Macro F1 = 0.9665

3. cells – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 3
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 8
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.4373 - Train Acc: 84.42% | Val Loss: 0.7451 - Val Acc: 79.44%
Epoch [02/15] | Train Loss: 0.2135 - Train Acc: 92.99% | Val Loss: 1.0535 - Val Acc: 73.45%
Epoch [03/15] | Train Loss: 0.1472 - Train Acc: 95.23% | Val Loss: 0.4059 - Val Acc: 85.52%
Epoch [04/15] | Train Loss: 0.1450 - Train Acc: 95.02% | Val Loss: 0.2614 - Val Acc: 90.71%
Epoch [05/15] | Train Loss: 0.1137 - Train Acc: 96.25% | Val Loss: 0.7099 - Val Acc: 80.61%
Epoch [06/15] | Train Loss: 0.1011 - Train Acc: 96.54% | Val Loss: 0.2834 - Val Acc: 90.86%
Epoch [07/15] | Train Loss: 0.1131 - Train Acc: 96.21% | Val Loss: 0.2845 - Val Acc: 90.56%
Epoch [08/15] | Train Loss: 0.0874 - Train Acc: 97.11% | Val Loss: 0.3889 - Val Acc: 88.88%
Epoch [09/15] | Train Loss: 0.0906 - Train Acc: 96.68% | Val Loss: 0.4331 - Val Acc: 86.47%
Epoch [10/15] | Train Loss: 0.0812 - Train Acc: 97.24% | Val Loss: 0.3649 - Val Acc: 88.95%
Epoch [11/15] | Train Loss: 0.0669 - Train Acc: 97.71% | Val Loss: 0.5524 - Val Acc: 86.10%
Epoch [12/15] | Train Loss: 0.0813 - Train Acc: 97.22% | Val Loss: 0.5019 - Val Acc: 85.81%
Epoch [13/15] | Train Loss: 0.0628 - Train Acc: 97.85% | Val Loss: 0.6183 - Val Acc: 85.59%
Epoch [14/15] | Train Loss: 0.0668 - Train Acc: 97.74% | Val Loss: 0.2208 - Val Acc: 93.20%
Epoch [15/15] | Train Loss: 0.0569 - Train Acc: 98.07% | Val Loss: 0.1266 - Val Acc: 96.12%
Best checkpoint saved: best_model/cells_ResNet18.pth (epoch=15, val_acc=96.12%)
Training Complete!
```

Final Test Results (ResNet18 on cells): Accuracy = 96.35% | Precision (macro) = 0.9599 | Recall (macro) = 0.9641 | Macro F1 = 0.9613

Dataset Used – chest

Chest X-ray (2 classes, grayscale) | Expected minimum test accuracy: 87%

4. chest – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 1
CRITERION.................... CrossEntropyLoss
```

```
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 2
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 1e-4
```

Logs:

```
Epoch [01/15] | Train Loss: 0.3124 - Train Acc: 87.15% | Val Loss: 0.2966 - Val Acc: 91.97%
Epoch [02/15] | Train Loss: 0.1586 - Train Acc: 94.20% | Val Loss: 0.2135 - Val Acc: 94.84%
Epoch [03/15] | Train Loss: 0.1320 - Train Acc: 94.84% | Val Loss: 0.1117 - Val Acc: 95.22%
Epoch [04/15] | Train Loss: 0.1228 - Train Acc: 95.63% | Val Loss: 0.1242 - Val Acc: 95.79%
Epoch [05/15] | Train Loss: 0.0795 - Train Acc: 97.01% | Val Loss: 0.4370 - Val Acc: 86.04%
Epoch [06/15] | Train Loss: 0.0965 - Train Acc: 96.56% | Val Loss: 0.3662 - Val Acc: 86.62%
Epoch [07/15] | Train Loss: 0.0977 - Train Acc: 96.58% | Val Loss: 0.1532 - Val Acc: 95.03%
Epoch [08/15] | Train Loss: 0.0780 - Train Acc: 97.41% | Val Loss: 0.1836 - Val Acc: 95.79%
Epoch [09/15] | Train Loss: 0.0614 - Train Acc: 97.88% | Val Loss: 0.1328 - Val Acc: 95.98%
Epoch [10/15] | Train Loss: 0.0595 - Train Acc: 97.90% | Val Loss: 0.1617 - Val Acc: 95.03%
Epoch [11/15] | Train Loss: 0.0482 - Train Acc: 98.19% | Val Loss: 0.1559 - Val Acc: 95.22%
Epoch [12/15] | Train Loss: 0.0465 - Train Acc: 98.45% | Val Loss: 0.0932 - Val Acc: 96.75%
Epoch [13/15] | Train Loss: 0.0471 - Train Acc: 98.51% | Val Loss: 0.1422 - Val Acc: 95.98%
Epoch [14/15] | Train Loss: 0.0393 - Train Acc: 98.47% | Val Loss: 0.0817 - Val Acc: 97.90%
Epoch [15/15] | Train Loss: 0.0348 - Train Acc: 98.92% | Val Loss: 0.1391 - Val Acc: 95.22%
Best checkpoint saved: best_model/chest_AlexNet.pth (epoch=14, val_acc=97.90%)
Training Complete!
```

Final Test Results (AlexNet on chest): Accuracy = 86.38% | Precision (macro) = 0.9054 | Recall (macro) = 0.8201 | Macro F1 = 0.8408

5. chest – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 1
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 2
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 1e-4
```

Logs:

```
Epoch [01/15] | Train Loss: 0.2293 - Train Acc: 90.59% | Val Loss: 0.3932 - Val Acc: 85.85%
Epoch [02/15] | Train Loss: 0.1342 - Train Acc: 95.29% | Val Loss: 0.5343 - Val Acc: 83.17%
Epoch [03/15] | Train Loss: 0.0983 - Train Acc: 96.45% | Val Loss: 0.1144 - Val Acc: 95.22%
Epoch [04/15] | Train Loss: 0.0907 - Train Acc: 96.58% | Val Loss: 0.1326 - Val Acc: 95.60%
Epoch [05/15] | Train Loss: 0.0853 - Train Acc: 96.62% | Val Loss: 1.8916 - Val Acc: 61.19%
Epoch [06/15] | Train Loss: 0.1042 - Train Acc: 96.50% | Val Loss: 0.1755 - Val Acc: 95.98%
Epoch [07/15] | Train Loss: 0.0813 - Train Acc: 97.15% | Val Loss: 0.1425 - Val Acc: 94.65%
Epoch [08/15] | Train Loss: 0.0695 - Train Acc: 97.49% | Val Loss: 0.0914 - Val Acc: 95.60%
Epoch [09/15] | Train Loss: 0.0548 - Train Acc: 97.98% | Val Loss: 0.2228 - Val Acc: 95.03%
Epoch [10/15] | Train Loss: 0.0496 - Train Acc: 98.15% | Val Loss: 0.0964 - Val Acc: 95.98%
Epoch [11/15] | Train Loss: 0.0610 - Train Acc: 97.90% | Val Loss: 0.3047 - Val Acc: 91.20%
Epoch [12/15] | Train Loss: 0.0487 - Train Acc: 98.13% | Val Loss: 0.1201 - Val Acc: 96.18%
Epoch [13/15] | Train Loss: 0.0488 - Train Acc: 98.24% | Val Loss: 0.1335 - Val Acc: 95.98%
Epoch [14/15] | Train Loss: 0.0437 - Train Acc: 98.81% | Val Loss: 0.2581 - Val Acc: 95.98%
Epoch [15/15] | Train Loss: 0.0345 - Train Acc: 98.68% | Val Loss: 0.0953 - Val Acc: 96.94%
Best checkpoint saved: best_model/chest_VGG16.pth (epoch=15, val_acc=96.94%)
Training Complete!
```

Final Test Results (VGG16 on chest): Accuracy = 82.21% | Precision (macro) = 0.8892 | Recall (macro) = 0.7628 | Macro F1 = 0.7822

6. chest – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 1
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
EPOCHS...................... 15
LEARNING_RATE............... 0.001
NUM_CLASSES................. 2
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 1e-4
```

Logs:

```
Epoch [01/15] | Train Loss: 0.2396 - Train Acc: 90.25% | Val Loss: 0.3194 - Val Acc: 87.95%
Epoch [02/15] | Train Loss: 0.1516 - Train Acc: 94.18% | Val Loss: 0.4163 - Val Acc: 86.81%
Epoch [03/15] | Train Loss: 0.1176 - Train Acc: 95.52% | Val Loss: 0.2554 - Val Acc: 91.59%
Epoch [04/15] | Train Loss: 0.1124 - Train Acc: 95.82% | Val Loss: 0.1806 - Val Acc: 93.69%
Epoch [05/15] | Train Loss: 0.0987 - Train Acc: 96.24% | Val Loss: 1.6548 - Val Acc: 65.97%
Epoch [06/15] | Train Loss: 0.0926 - Train Acc: 96.64% | Val Loss: 0.2739 - Val Acc: 91.01%
Epoch [07/15] | Train Loss: 0.0781 - Train Acc: 97.11% | Val Loss: 0.1037 - Val Acc: 95.98%
Epoch [08/15] | Train Loss: 0.0702 - Train Acc: 97.52% | Val Loss: 0.1801 - Val Acc: 93.69%
Epoch [09/15] | Train Loss: 0.0729 - Train Acc: 97.09% | Val Loss: 0.1435 - Val Acc: 93.88%
Epoch [10/15] | Train Loss: 0.0638 - Train Acc: 97.83% | Val Loss: 0.1197 - Val Acc: 95.79%
Epoch [11/15] | Train Loss: 0.0601 - Train Acc: 97.81% | Val Loss: 0.2391 - Val Acc: 92.35%
Epoch [12/15] | Train Loss: 0.0569 - Train Acc: 97.81% | Val Loss: 0.1293 - Val Acc: 95.60%
Epoch [13/15] | Train Loss: 0.0540 - Train Acc: 98.00% | Val Loss: 0.2655 - Val Acc: 91.59%
Epoch [14/15] | Train Loss: 0.0507 - Train Acc: 98.07% | Val Loss: 0.7436 - Val Acc: 81.64%
Epoch [15/15] | Train Loss: 0.0358 - Train Acc: 98.68% | Val Loss: 1.1647 - Val Acc: 78.20%
Best checkpoint saved: best_model/chest_ResNet18.pth (epoch=7, val_acc=95.98%)
Training Complete!
```

Final Test Results (ResNet18 on chest): Accuracy = 84.46% | Precision (macro) = 0.8866 | Recall (macro) = 0.7970 | Macro F1 = 0.8168

Dataset Used – lesions

Skin lesion dermatoscopy (7 classes, RGB) | Expected minimum test accuracy: 67%

7. lesions – AlexNet

Total epochs trained: 25

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 25
LEARNING_RATE............... 0.001
NUM_CLASSES................. 7
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 1.0529 - Train Acc: 65.61% | Val Loss: 1.1957 - Val Acc: 70.16%
Epoch [02/25] | Train Loss: 0.8855 - Train Acc: 67.90% | Val Loss: 0.8012 - Val Acc: 70.41%
Epoch [03/25] | Train Loss: 0.8424 - Train Acc: 70.00% | Val Loss: 0.7480 - Val Acc: 73.41%
Epoch [04/25] | Train Loss: 0.8165 - Train Acc: 70.72% | Val Loss: 0.9198 - Val Acc: 63.05%
Epoch [05/25] | Train Loss: 0.8063 - Train Acc: 70.76% | Val Loss: 0.8240 - Val Acc: 71.16%
Epoch [06/25] | Train Loss: 0.7798 - Train Acc: 72.01% | Val Loss: 0.6731 - Val Acc: 75.53%
Epoch [07/25] | Train Loss: 0.7447 - Train Acc: 72.62% | Val Loss: 0.6804 - Val Acc: 74.78%
Epoch [08/25] | Train Loss: 0.7338 - Train Acc: 73.69% | Val Loss: 0.6499 - Val Acc: 75.91%
Epoch [09/25] | Train Loss: 0.7204 - Train Acc: 73.98% | Val Loss: 0.6923 - Val Acc: 74.66%
Epoch [10/25] | Train Loss: 0.7095 - Train Acc: 74.60% | Val Loss: 0.6509 - Val Acc: 76.78%
Epoch [11/25] | Train Loss: 0.6972 - Train Acc: 74.38% | Val Loss: 0.7455 - Val Acc: 73.41%
Epoch [12/25] | Train Loss: 0.6763 - Train Acc: 75.61% | Val Loss: 0.6667 - Val Acc: 75.28%
Epoch [13/25] | Train Loss: 0.6642 - Train Acc: 75.68% | Val Loss: 0.8356 - Val Acc: 72.53%
Epoch [14/25] | Train Loss: 0.6372 - Train Acc: 76.40% | Val Loss: 0.9236 - Val Acc: 60.67%
Epoch [15/25] | Train Loss: 0.6312 - Train Acc: 76.95% | Val Loss: 1.0745 - Val Acc: 55.93%
Epoch [16/25] | Train Loss: 0.6127 - Train Acc: 77.20% | Val Loss: 0.6772 - Val Acc: 75.28%
Epoch [17/25] | Train Loss: 0.6067 - Train Acc: 77.76% | Val Loss: 0.6645 - Val Acc: 76.65%
```

```
Epoch [18/25] | Train Loss: 0.5895 - Train Acc: 78.18% | Val Loss: 0.6799 - Val Acc: 74.28%
Epoch [19/25] | Train Loss: 0.5719 - Train Acc: 78.68% | Val Loss: 0.9693 - Val Acc: 72.53%
Epoch [20/25] | Train Loss: 0.5632 - Train Acc: 79.40% | Val Loss: 0.6156 - Val Acc: 78.28%
Epoch [21/25] | Train Loss: 0.5479 - Train Acc: 80.08% | Val Loss: 0.6048 - Val Acc: 78.78%
Epoch [22/25] | Train Loss: 0.5367 - Train Acc: 80.25% | Val Loss: 1.0047 - Val Acc: 74.78%
Epoch [23/25] | Train Loss: 0.5145 - Train Acc: 81.59% | Val Loss: 1.0500 - Val Acc: 62.92%
Epoch [24/25] | Train Loss: 0.5147 - Train Acc: 81.05% | Val Loss: 1.1849 - Val Acc: 58.55%
Epoch [25/25] | Train Loss: 0.5011 - Train Acc: 81.45% | Val Loss: 0.6626 - Val Acc: 77.15%
Best checkpoint saved: best_model/lesions_AlexNet.pth (epoch=21, val_acc=78.78%)
Training Complete!
```

Final Test Results (AlexNet on lesions): Accuracy = 76.01% | Precision (macro) = 0.5651 | Recall (macro) = 0.4770 | Macro F1 = 0.4786

8. lesions – VGG16

Total epochs trained: 25

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 25
LEARNING_RATE............... 0.001
NUM_CLASSES................. 7
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 1.0747 - Train Acc: 65.63% | Val Loss: 0.9127 - Val Acc: 70.16%
Epoch [02/25] | Train Loss: 0.9660 - Train Acc: 66.60% | Val Loss: 0.8620 - Val Acc: 69.29%
Epoch [03/25] | Train Loss: 0.9424 - Train Acc: 66.32% | Val Loss: 0.8262 - Val Acc: 68.54%
Epoch [04/25] | Train Loss: 0.9072 - Train Acc: 67.01% | Val Loss: 0.8024 - Val Acc: 70.79%
Epoch [05/25] | Train Loss: 0.9077 - Train Acc: 66.83% | Val Loss: 0.7764 - Val Acc: 71.66%
Epoch [06/25] | Train Loss: 0.8819 - Train Acc: 67.62% | Val Loss: 0.7903 - Val Acc: 71.29%
Epoch [07/25] | Train Loss: 0.8882 - Train Acc: 67.43% | Val Loss: 0.8117 - Val Acc: 69.29%
Epoch [08/25] | Train Loss: 0.8679 - Train Acc: 68.01% | Val Loss: 0.7641 - Val Acc: 71.91%
Epoch [09/25] | Train Loss: 0.8726 - Train Acc: 68.37% | Val Loss: 0.8375 - Val Acc: 67.04%
Epoch [10/25] | Train Loss: 0.8446 - Train Acc: 69.64% | Val Loss: 0.7748 - Val Acc: 72.78%
Epoch [11/25] | Train Loss: 0.8310 - Train Acc: 69.57% | Val Loss: 0.7272 - Val Acc: 74.28%
Epoch [12/25] | Train Loss: 0.8144 - Train Acc: 69.82% | Val Loss: 0.7338 - Val Acc: 73.53%
Epoch [13/25] | Train Loss: 0.8125 - Train Acc: 70.27% | Val Loss: 0.7923 - Val Acc: 70.29%
Epoch [14/25] | Train Loss: 0.8022 - Train Acc: 70.50% | Val Loss: 0.8496 - Val Acc: 69.04%
Epoch [15/25] | Train Loss: 0.7924 - Train Acc: 70.61% | Val Loss: 0.7144 - Val Acc: 72.91%
Epoch [16/25] | Train Loss: 0.7932 - Train Acc: 70.61% | Val Loss: 0.9979 - Val Acc: 62.17%
Epoch [17/25] | Train Loss: 0.7909 - Train Acc: 70.76% | Val Loss: 0.6833 - Val Acc: 73.66%
Epoch [18/25] | Train Loss: 0.7714 - Train Acc: 71.47% | Val Loss: 0.7171 - Val Acc: 72.28%
Epoch [19/25] | Train Loss: 0.7749 - Train Acc: 71.79% | Val Loss: 0.7009 - Val Acc: 75.16%
Epoch [20/25] | Train Loss: 0.7584 - Train Acc: 71.83% | Val Loss: 1.5402 - Val Acc: 56.05%
Epoch [21/25] | Train Loss: 0.7425 - Train Acc: 73.23% | Val Loss: 0.8270 - Val Acc: 69.79%
Epoch [22/25] | Train Loss: 0.7476 - Train Acc: 72.51% | Val Loss: 0.7117 - Val Acc: 73.16%
Epoch [23/25] | Train Loss: 0.7246 - Train Acc: 73.10% | Val Loss: 0.8416 - Val Acc: 74.41%
Epoch [24/25] | Train Loss: 0.7221 - Train Acc: 73.57% | Val Loss: 0.6949 - Val Acc: 75.03%
Epoch [25/25] | Train Loss: 0.7133 - Train Acc: 74.27% | Val Loss: 0.8404 - Val Acc: 67.29%
Best checkpoint saved: best_model/lesions_VGG16.pth (epoch=19, val_acc=75.16%)
Training Complete!
[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).
```

Final Test Results (VGG16 on lesions): Accuracy = 71.22% | Precision (macro) = 0.3895 | Recall (macro) = 0.3053 | Macro F1 = 0.3239

9. lesions – ResNet18

Total epochs trained: 25

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
EPOCHS...................... 25
LEARNING_RATE............... 0.001
```

```
NUM_CLASSES.................. 7
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 0.9720 - Train Acc: 66.03% | Val Loss: 0.9265 - Val Acc: 68.41%
Epoch [02/25] | Train Loss: 0.8537 - Train Acc: 68.29% | Val Loss: 0.7988 - Val Acc: 71.79%
Epoch [03/25] | Train Loss: 0.8348 - Train Acc: 69.14% | Val Loss: 0.7086 - Val Acc: 74.03%
Epoch [04/25] | Train Loss: 0.8075 - Train Acc: 70.56% | Val Loss: 0.8540 - Val Acc: 63.92%
Epoch [05/25] | Train Loss: 0.7779 - Train Acc: 70.69% | Val Loss: 0.7274 - Val Acc: 72.53%
Epoch [06/25] | Train Loss: 0.7519 - Train Acc: 72.01% | Val Loss: 0.7053 - Val Acc: 73.16%
Epoch [07/25] | Train Loss: 0.7354 - Train Acc: 72.51% | Val Loss: 0.7191 - Val Acc: 72.16%
Epoch [08/25] | Train Loss: 0.7096 - Train Acc: 73.64% | Val Loss: 0.9580 - Val Acc: 67.29%
Epoch [09/25] | Train Loss: 0.7143 - Train Acc: 73.24% | Val Loss: 0.7481 - Val Acc: 73.03%
Epoch [10/25] | Train Loss: 0.6814 - Train Acc: 74.85% | Val Loss: 0.8472 - Val Acc: 69.16%
Epoch [11/25] | Train Loss: 0.6762 - Train Acc: 75.35% | Val Loss: 0.7199 - Val Acc: 71.41%
Epoch [12/25] | Train Loss: 0.6628 - Train Acc: 75.35% | Val Loss: 0.6290 - Val Acc: 76.65%
Epoch [13/25] | Train Loss: 0.6530 - Train Acc: 75.92% | Val Loss: 0.7214 - Val Acc: 72.41%
Epoch [14/25] | Train Loss: 0.6519 - Train Acc: 75.52% | Val Loss: 0.6432 - Val Acc: 76.03%
Epoch [15/25] | Train Loss: 0.6401 - Train Acc: 76.24% | Val Loss: 0.7095 - Val Acc: 72.41%
Epoch [16/25] | Train Loss: 0.6296 - Train Acc: 76.72% | Val Loss: 0.6952 - Val Acc: 74.78%
Epoch [17/25] | Train Loss: 0.6165 - Train Acc: 76.96% | Val Loss: 0.6660 - Val Acc: 77.03%
Epoch [18/25] | Train Loss: 0.6096 - Train Acc: 76.95% | Val Loss: 0.6209 - Val Acc: 77.15%
Epoch [19/25] | Train Loss: 0.5981 - Train Acc: 77.86% | Val Loss: 0.6167 - Val Acc: 77.65%
Epoch [20/25] | Train Loss: 0.5887 - Train Acc: 77.89% | Val Loss: 0.8091 - Val Acc: 70.79%
Epoch [21/25] | Train Loss: 0.5802 - Train Acc: 78.30% | Val Loss: 0.6184 - Val Acc: 78.53%
Epoch [22/25] | Train Loss: 0.5553 - Train Acc: 79.16% | Val Loss: 0.6607 - Val Acc: 74.91%
Epoch [23/25] | Train Loss: 0.5361 - Train Acc: 79.94% | Val Loss: 0.6097 - Val Acc: 77.78%
Epoch [24/25] | Train Loss: 0.5393 - Train Acc: 79.62% | Val Loss: 0.7076 - Val Acc: 75.16%
Epoch [25/25] | Train Loss: 0.5107 - Train Acc: 81.05% | Val Loss: 0.9810 - Val Acc: 63.80%
Best checkpoint saved: best_model/lesions_ResNet18.pth (epoch=21, val_acc=78.53%)
Training Complete!
[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).
```

Final Test Results (ResNet18 on lesions): Accuracy = 76.46% | Precision (macro) = 0.6136 | Recall (macro) = 0.4205 | Macro F1 = 0.4719

Dataset Used – orgs

Organ images (11 classes, grayscale) | Expected minimum test accuracy: 83%

10. orgs – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 1
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 11
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.8939 - Train Acc: 68.87% | Val Loss: 0.4486 - Val Acc: 82.88%
Epoch [02/15] | Train Loss: 0.3669 - Train Acc: 88.23% | Val Loss: 0.4018 - Val Acc: 87.43%
Epoch [03/15] | Train Loss: 0.2964 - Train Acc: 90.30% | Val Loss: 0.4067 - Val Acc: 87.76%
Epoch [04/15] | Train Loss: 0.2334 - Train Acc: 92.26% | Val Loss: 0.1587 - Val Acc: 94.27%
Epoch [05/15] | Train Loss: 0.2069 - Train Acc: 93.15% | Val Loss: 0.2430 - Val Acc: 92.25%
Epoch [06/15] | Train Loss: 0.1724 - Train Acc: 94.26% | Val Loss: 0.2111 - Val Acc: 91.86%
Epoch [07/15] | Train Loss: 0.1439 - Train Acc: 95.13% | Val Loss: 0.1677 - Val Acc: 94.86%
Epoch [08/15] | Train Loss: 0.1488 - Train Acc: 94.99% | Val Loss: 0.1683 - Val Acc: 94.14%
Epoch [09/15] | Train Loss: 0.1360 - Train Acc: 95.57% | Val Loss: 0.3312 - Val Acc: 89.06%
Epoch [10/15] | Train Loss: 0.1337 - Train Acc: 95.65% | Val Loss: 0.0980 - Val Acc: 96.03%
Epoch [11/15] | Train Loss: 0.1275 - Train Acc: 95.78% | Val Loss: 0.2609 - Val Acc: 94.08%
Epoch [12/15] | Train Loss: 0.1035 - Train Acc: 96.56% | Val Loss: 0.5707 - Val Acc: 87.37%
Epoch [13/15] | Train Loss: 0.1141 - Train Acc: 96.32% | Val Loss: 1.6973 - Val Acc: 77.60%
Epoch [14/15] | Train Loss: 0.1044 - Train Acc: 96.55% | Val Loss: 0.2073 - Val Acc: 92.90%
Epoch [15/15] | Train Loss: 0.1002 - Train Acc: 96.94% | Val Loss: 0.1662 - Val Acc: 94.21%
```

```
Best checkpoint saved: best_model/orgs_AlexNet.pth (epoch=10, val_acc=96.03%)
Training Complete!
```

Final Test Results (AlexNet on orgs): Accuracy = 90.07% | Precision (macro) = 0.8894 | Recall (macro) = 0.8856 | Macro F1 = 0.8855

11. orgs – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 1
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 15
LEARNING_RATE............... 0.001
NUM_CLASSES................. 11
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 0
```

Logs:

```
Epoch [01/15] | Train Loss: 1.3792 - Train Acc: 49.57% | Val Loss: 1.0464 - Val Acc: 61.65%
Epoch [02/15] | Train Loss: 0.6527 - Train Acc: 75.22% | Val Loss: 0.6058 - Val Acc: 78.58%
Epoch [03/15] | Train Loss: 0.5041 - Train Acc: 80.48% | Val Loss: 0.3726 - Val Acc: 84.05%
Epoch [04/15] | Train Loss: 0.4022 - Train Acc: 84.26% | Val Loss: 0.3266 - Val Acc: 86.52%
Epoch [05/15] | Train Loss: 0.3426 - Train Acc: 87.14% | Val Loss: 0.3445 - Val Acc: 86.00%
Epoch [06/15] | Train Loss: 0.3499 - Train Acc: 87.55% | Val Loss: 0.2428 - Val Acc: 90.56%
Epoch [07/15] | Train Loss: 0.2900 - Train Acc: 89.56% | Val Loss: 0.2845 - Val Acc: 89.97%
Epoch [08/15] | Train Loss: 0.2303 - Train Acc: 91.90% | Val Loss: 0.3085 - Val Acc: 89.52%
Epoch [09/15] | Train Loss: 0.2082 - Train Acc: 93.23% | Val Loss: 0.3986 - Val Acc: 88.28%
Epoch [10/15] | Train Loss: 0.3449 - Train Acc: 89.00% | Val Loss: 0.1844 - Val Acc: 93.55%
Epoch [11/15] | Train Loss: 0.1889 - Train Acc: 93.63% | Val Loss: 0.1514 - Val Acc: 94.60%
Epoch [12/15] | Train Loss: 0.1877 - Train Acc: 93.93% | Val Loss: 0.1524 - Val Acc: 94.66%
Epoch [13/15] | Train Loss: 0.1630 - Train Acc: 94.74% | Val Loss: 0.1758 - Val Acc: 94.14%
Epoch [14/15] | Train Loss: 0.1272 - Train Acc: 95.72% | Val Loss: 0.1396 - Val Acc: 95.44%
Epoch [15/15] | Train Loss: 0.1163 - Train Acc: 96.31% | Val Loss: 0.1480 - Val Acc: 94.79%
Best checkpoint saved: best_model/orgs_VGG16.pth (epoch=14, val_acc=95.44%)
Training Complete!
```

Final Test Results (VGG16 on orgs): Accuracy = 88.90% | Precision (macro) = 0.8812 | Recall (macro) = 0.8691 | Macro F1 = 0.8722

12. orgs – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 1
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
EPOCHS...................... 15
LEARNING_RATE............... 0.001
NUM_CLASSES................. 11
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.6646 - Train Acc: 75.92% | Val Loss: 0.5742 - Val Acc: 81.32%
Epoch [02/15] | Train Loss: 0.3328 - Train Acc: 88.06% | Val Loss: 0.6068 - Val Acc: 80.21%
Epoch [03/15] | Train Loss: 0.2415 - Train Acc: 91.37% | Val Loss: 0.1964 - Val Acc: 92.90%
Epoch [04/15] | Train Loss: 0.1867 - Train Acc: 93.49% | Val Loss: 0.2044 - Val Acc: 92.84%
Epoch [05/15] | Train Loss: 0.1594 - Train Acc: 94.33% | Val Loss: 0.2323 - Val Acc: 91.54%
Epoch [06/15] | Train Loss: 0.1257 - Train Acc: 95.46% | Val Loss: 0.2882 - Val Acc: 91.08%
Epoch [07/15] | Train Loss: 0.1260 - Train Acc: 95.52% | Val Loss: 0.1832 - Val Acc: 93.95%
Epoch [08/15] | Train Loss: 0.0910 - Train Acc: 96.81% | Val Loss: 0.1936 - Val Acc: 92.90%
Epoch [09/15] | Train Loss: 0.1117 - Train Acc: 96.02% | Val Loss: 0.0998 - Val Acc: 96.16%
Epoch [10/15] | Train Loss: 0.0885 - Train Acc: 97.01% | Val Loss: 0.1554 - Val Acc: 94.99%
Epoch [11/15] | Train Loss: 0.0644 - Train Acc: 97.75% | Val Loss: 0.1134 - Val Acc: 95.96%
Epoch [12/15] | Train Loss: 0.0489 - Train Acc: 98.25% | Val Loss: 0.1877 - Val Acc: 94.08%
```

```
Epoch [13/15] | Train Loss: 0.0668 - Train Acc: 97.67% | Val Loss: 0.0999 - Val Acc: 96.09%
Epoch [14/15] | Train Loss: 0.0460 - Train Acc: 98.45% | Val Loss: 0.0729 - Val Acc: 97.14%
Epoch [15/15] | Train Loss: 0.0408 - Train Acc: 98.58% | Val Loss: 0.1739 - Val Acc: 94.79%
Best checkpoint saved: best_model/orgs_ResNet18.pth (epoch=14, val_acc=97.14%)
Training Complete!
```

Final Test Results (ResNet18 on orgs): Accuracy = 92.00% | Precision (macro) = 0.9109 | Recall (macro) = 0.9088 | Macro F1 = 0.9078

Consolidated Summary – All Permutations

Dataset	Model	Accuracy (%)	Precision	Recall	Macro F1
cells	AlexNet	95.15	0.947	0.944	0.945
cells	VGG16	96.78	0.965	0.969	0.967
cells	ResNet18	96.35	0.960	0.964	0.961
chest	AlexNet	86.38	0.905	0.820	0.841
chest	VGG16	82.21	0.889	0.763	0.782
chest	ResNet18	84.46	0.887	0.797	0.817
lesions	AlexNet	76.01	0.565	0.477	0.479
lesions	VGG16	71.22	0.390	0.305	0.324
lesions	ResNet18	76.46	0.614	0.420	0.472
orgs	AlexNet	90.07	0.889	0.886	0.885
orgs	VGG16	88.90	0.881	0.869	0.872
orgs	ResNet18	92.00	0.911	0.909	0.908

Architectural Recommendations

Based on the benchmark results across all twelve dataset–model permutations, the recommended pairings are:

Dataset	Recommended model	Test accuracy	Macro F1	Rationale
cells	VGG16	96.78%	0.967	Highest accuracy and F1; AlexNet (95.15%) is a strong lighter-weight alternative when compute is constrained.
chest	AlexNet	86.38%	0.841	Best accuracy and F1 of the three; the shallow architecture generalizes better on this small 2-class set, while the deeper networks overfit.
lesions	ResNet18	76.46%	0.472	Highest accuracy; residual connections help on this hard, imbalanced 7-class task. AlexNet is comparable (76.01%) with a marginally higher F1.
orgs	ResNet18	92.00%	0.908	Best accuracy, precision, recall, and F1 across the board.

Summary: ResNet18 is the strongest general-purpose choice, winning on the two hardest datasets (lesions and orgs) and coming within 0.4% of the best on cells. VGG16 edges out the others only on cells, but at a much higher computational cost. AlexNet is the clear pick for chest and remains a competitive, efficient option everywhere, making it the best accuracy-per-compute trade-off when resources are limited.

Part 2 - Green Initiative: Architectural Downscaling and Efficiency Analysis

Objective

The original model registry (AlexNet, VGG16, ResNet18) relies on heavy, over-parameterized topologies. The goal of this phase was to redesign these architectures to drastically reduce computational and energy cost while preserving baseline classification accuracy, and to quantify the trade-off across every dataset-model permutation.

Architectural changes

The redesign targeted the largest sources of parameters and compute in each network. For AlexNet and VGG16, the fully-connected classifier heads (which held the vast majority of parameters) were replaced with a single global average pooling (GAP) layer followed by one linear layer; VGG16 was additionally shortened from five convolutional blocks to three. For ResNet18, each stage was reduced from two residual blocks to one. All models retain the same (in_channels, num_classes) interface and were trained and evaluated under identical settings (epochs, batch size, optimizer, hardware) as the baseline, so the comparison is fair.

Efficiency Verification Matrix

Every dataset-model run now also logs training runtime, inference latency per sample, and peak memory for both training and inference. Values below are shown as original to slimmed.

Dataset	Model	Params (M)	Train time (s)	Inference latency (ms/sample)	Peak mem train / infer (MB)
cells	AlexNet	5.69 to 1.49	58 to 51	0.074 to 0.071	185 to 118 / 132 to 66
cells	VGG16	12.63 to 1.22	315 to 227	0.494 to 0.365	878 to 619 / 410 to 231
cells	ResNet18	11.17 to 4.90	663 to 351	1.008 to 0.528	1339 to 783 / 516 to 351
chest	AlexNet	5.68 to 1.48	21 to 18	0.070 to 0.069	229 to 117 / 177 to 64
chest	VGG16	12.63 to 1.21	121 to 86	0.481 to 0.362	873 to 616 / 406 to 229
chest	ResNet18	11.17 to 4.90	251 to 133	1.004 to 0.525	1337 to 781 / 512 to 350
lesions	AlexNet	5.69 to 1.49	55 to 48	0.076 to 0.072	234 to 120 / 180 to 67
lesions	VGG16	12.63 to 1.22	309 to 224	0.494 to 0.366	874 to 618 / 407 to 230
lesions	ResNet18	11.17 to 4.90	646 to 344	1.013 to 0.530	1338 to 783 / 514 to 351
orgs	AlexNet	5.69 to 1.49	60 to 52	0.070 to 0.068	229 to 117 / 177 to 64
orgs	VGG16	12.63 to 1.22	354 to 254	0.481 to 0.361	875 to 616 / 409 to 229
orgs	ResNet18	11.17 to 4.90	739 to 391	1.008 to 0.524	1339 to 782 / 513 to 350

Across architectures the parameter reduction is ~74% (AlexNet), ~90% (VGG16), and ~56% (ResNet18). Inference latency falls ~26% for VGG16 and ~48% for ResNet18 (AlexNet was already minimal), peak inference memory drops 32-64%, and training runtime drops up to ~47%.

Accuracy preservation

Dataset	Model	Accuracy (%)	Macro F1
cells	AlexNet	95.5 to 94.9	0.950 to 0.944
cells	VGG16	95.4 to 97.5	0.951 to 0.972
cells	ResNet18	97.5 to 97.1	0.975 to 0.968
chest	AlexNet	85.1 to 88.5	0.823 to 0.867
chest	VGG16	85.3 to 86.1	0.825 to 0.837
chest	ResNet18	78.5 to 81.6	0.727 to 0.774
lesions	AlexNet	75.4 to 75.6	0.477 to 0.474
lesions	VGG16	73.3 to 77.1	0.423 to 0.502
lesions	ResNet18	76.2 to 75.4	0.457 to 0.456
orgs	AlexNet	89.4 to 91.0	0.881 to 0.899
orgs	VGG16	88.4 to 92.7	0.870 to 0.917
orgs	ResNet18	91.5 to 92.6	0.903 to 0.916

Trade-off discussion

The slimmed models matched or beat the original accuracy while using far less compute. VGG16 shows this best: its parameters dropped by about 90% (12.6M to 1.2M), yet its accuracy went up on every dataset. This proves the original fully-connected head was wasted capacity for these small 64x64 images. The baseline models were simply too large for datasets of this size, so cutting them down lost no useful power. Swapping the dense head for global average pooling also removed the layers most likely to memorize the training data, which slightly improved generalization.

Recommendation

The green-optimized architectures are the recommended choice for deployment on diagnostic devices: they deliver equal or better accuracy while cutting parameters by 56-90%, inference latency by up to ~48%, and memory footprint by 30-64%. For future work, adding a learning rate scheduler would stabilize the validation curves, and class-balancing or augmentation on lesions would further improve the model.